

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING

Kathmandu Engineering College
Department of Computer Engineering



Mid-term Report

On

SIKSHYALAYA

Empowering Education, Hand in Hand with Tradition

[Code No: CT 755]

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ABSTRACT

Sikshalaya is a mobile application that aims to change the way student study by providing them with an exceptional learning experience. With its user-friendly interface and a variety of features, Sikshalaya offers a wide range of educational resources to enhance learning. This innovative platform provides students with an extensive range of educational tools, including note access as PDF format, offline video streaming with download availability, and real time assessments through live tests. Sikshalaya core mission is to deliver a user-centric, personalized learning experience, empowering students to access subject-specific materials such as lectures notes, videos, quizzes, and tests at their own pace via their smartphones. Using DotNet Core WebAPI for robust backend operations and seamless API integration, coupled with Angular for a user-friendly interface, Sikshalaya paves the way for an immersive and accessible e-learning environment.

Keywords: *Personalized, Innovative, Empowering, Immersive*

TABLE OF CONTENTS

ACKNOWLEDGEMENT.....	ii
ABSTRACT.....	iii
TABLE OF CONTENTS.....	iv
LIST OF FIGURES	vi
ABBREVIATION	vii
Chapter 1: INTRODUCTION.....	1
1.1 BACKGROUND THEORY	2
1.2 PROBLEM STATEMENT	3
1.3 OBJECTIVES.....	4
1.4 SCOPES AND APPLICATION.....	4
Chapter 2: LITERATURE REVIEW	6
2.1 EXISTING SYSTEMS.....	8
2.1.1 MeroSpark.....	8
2.1.2 BYJU's	8
2.1.3 MeroSchool.....	8
2.2 LIMITATIONS OF PREVIOUS SYSTEMS	8
2.3 SOLUTION PROPOSED BY OUR SYSTEM	9
Chapter 3: METHODOLOGY	10
3.1 PROCESS MODEL	10
3.2 SYSTEM BLOCK DIAGRAM	12
3.3 FLOWCHART	14
3.4 USE CASE DIAGRAM.....	15
3.5 OBJECT DIAGRAM.....	16
3.6 ACTIVITY DIAGRAM	17

3.7	DETAILED SEQUENCE DIAGRAM	18
3.8	DETAILED LOGIN SEQUENCE DIAGRAM	19
3.9	DETAILED TOPIC MODEL SEQUENCE DIAGRAM.....	19
3.10	STATE DIAGRAM	20
3.11	TOOLS USED	21
3.11.1	KOTLIN	21
3.11.2	ANDROID STUDIO	21
3.11.3	ANGULAR.....	22
3.11.4	DOT NET CORE WEBAPI	23
3.11.5	SWIFT.....	23
Chapter 4:	<i>EPILLOGUE</i>	25
4.1	TASK COMPLETED	25
4.2	TASK REMAINING.....	25
4.3	GANTT CHART.....	25
	<i>REFERENCES</i>	26
	<i>BIBLIOGRAPHY</i>	27
	<i>SCREENSHOTS</i>.....	28

LIST OF FIGURES

- Figure 3.1. Incremental Process Model
- Figure 3.2. System Block Diagram
- Figure 3.3. Flowchart of Application
- Figure 3.4. Use Case Diagram
- Figure 3.5. Object Diagram
- Figure 3.6. Activity Diagram
- Figure 3.7. Detailed Sequence Diagram
- Figure 3.8. Detailed Login Sequence Diagram
- Figure 3.9. Detailed Topic Model Sequence Diagram
- Figure 3.10. State Diagram
- Figure 4.3. Gantt Chart

ABBREVIATION

API	Application Programming Interface
E-Learning	Electronic Learning
IDE	Integrated Development Environment
LMS	Learning Management System
M-Learning	Mobile Learning
MOOC	Massive Open Online Course
PDF	Portable Document Format

Chapter 1: INTRODUCTION

In recent years, Nepal's education system has undergone a profound transformation propelled by technological advancements, with online learning emerging as a cornerstone of this evolution. With the advent of e-learning apps accessible on both Android and iOS devices, students across Nepal have found innovative ways to continue their education beyond the constraints of traditional classrooms, particularly in the wake of the challenges posed by the COVID-19 pandemic. These digital platforms offer a plethora of specialized content, ranging from interactive videos to live classes, effectively revolutionizing the learning experience and catering to individualized needs like never before.

At the forefront of this educational revolution stands the Mero School mobile app, a shining example of how technology is reshaping the educational landscape in Nepal. This user-friendly platform has empowered students with seamless access to course materials and facilitated collaborative discussions, transcending geographical barriers and fostering a sense of interconnectedness in the learning process. By embracing personalized learning over the archaic one-size-fits-all approach, teachers in Nepal are leveraging the power of technology to tailor education to the unique needs and learning styles of each student, thereby maximizing engagement and academic outcomes.

The significance of these e-learning apps extends far beyond merely providing an alternative to traditional classrooms. They represent a paradigm shift in education, ushering in an era of flexibility and accessibility previously unimaginable. By offering students the freedom to learn at their own pace, anytime and anywhere, these apps have democratized education, leveling the playing field and ensuring that no student is left behind. Moreover, the incorporation of technology not only facilitates a smoother transition post-pandemic but also heralds exciting possibilities for a more dynamic and engaging educational experience in Nepal.

As the educational landscape continues to evolve, it is evident that e-learning apps are poised to play an increasingly integral role in shaping the future of education in Nepal. With their ability to transcend physical boundaries and cater to diverse learning needs, these digital platforms hold the key to unlocking the full potential of every student, thereby laying the

foundation for a brighter and more prosperous future for generations to come. Embracing this technological revolution is not merely an option but a necessity if Nepal is to thrive in the ever-changing global landscape, where knowledge and innovation reign supreme.

1.1 BACKGROUND THEORY

The development of the Sikshalaya mobile application is driven by a profound recognition of the pivotal role smartphones play in the lives of students today. In an era where digital technology permeates every aspect of our daily existence, Sikshalaya aims to leverage the ubiquity and versatility of smartphones to revolutionize the landscape of education. Understanding the transformative potential of mobile technology in learning, Sikshalaya endeavors to harness this power to create a seamless and enriched learning experience for students across the globe.

At its core, Sikshalaya seeks to address the evolving needs and preferences of modern learners by providing them with a dynamic and user-centric platform for accessing educational resources. With the proliferation of digital content available across various online platforms, students often find themselves navigating a fragmented landscape, struggling to consolidate and organize disparate learning materials. Recognizing this challenge, Sikshalaya endeavors to streamline the learning process by offering a centralized hub where students can effortlessly access, manage, and engage with a wide array of educational content, all within the confines of a single mobile application.

The Sikshalaya mobile application is meticulously designed to simplify and expedite students' learning journeys through their smartphones. By integrating seamlessly with existing educational platforms and repositories, Sikshalaya ensures that students can conveniently access their course materials, textbooks, lecture notes, videos, and other resources from a diverse range of sources, all from the palm of their hand. Whether studying for exams, completing assignments, or conducting research, students can rely on Sikshalaya to provide them with instant access to the resources they need, anytime, anywhere.

Moreover, Sikshalaya is committed to fostering a supportive and engaging learning environment that encourages active participation and collaboration among students. Through features such as discussion forums, group chat functionalities, and collaborative projects,

Sikshalaya empowers students to connect with their peers, share insights, seek assistance, and collaborate on assignments, thereby fostering a sense of community and collective learning.

By embracing the power of mobile technology, Sikshalaya seeks to democratize access to education, transcending geographical barriers and socioeconomic constraints. Whether students reside in bustling urban centers or remote rural villages, Sikshalaya ensures that they have equitable access to high-quality educational resources and opportunities. In doing so, Sikshalaya not only enhances the educational experience of individual students but also contributes to the advancement of society as a whole.

In essence, Sikshalaya represents a paradigm shift in the way education is conceived, delivered, and experienced in the digital age. By harnessing the transformative potential of mobile technology, Sikshalaya empowers students to take ownership of their learning journey, personalize their educational experience, and unlock their full potential. In a world where knowledge is increasingly recognized as the ultimate currency, Sikshalaya stands as a beacon of innovation, accessibility, and empowerment in the realm of education.

1.2 PROBLEM STATEMENT

Students are facing difficulties in accessing effective online learning tools. The current e-learning platforms lack features that make it easy for students to work together on assignments or have meaningful discussions. The closure of schools and colleges due to events like strikes or disasters has made it clear that we need a better solution for learning from home. Additionally, not everyone has the same access to digital resources, creating a divide among students. Traditional methods also lack flexibility, making it difficult for learners to engage with content at their own pace, especially those with diverse schedules or work commitments. To solve these problems and improve education for everyone, we need to create a new e-learning app designed for Nepal. Developing an eLearning app aims to address these challenges by enhancing accessibility, offering flexible learning schedules, providing interactive features, and reducing costs associated with traditional educational methods. This app should be easy to use, help students work together on assignments, and make sure that all students, no matter where they are, can access quality education online.

1.3 OBJECTIVES

- To break down barriers to learning, making education more engaging and tailored to individual learners.

1.4 SCOPES AND APPLICATION

The project takes into consideration and covers the following areas:

- **DEVELOPMENT OF AN ONLINE LEARNING PLATFORM:** The development process involves creating a robust web application and mobile app that collectively serve as a comprehensive online learning platform. This platform becomes the digital space where students can not only access a vast array of educational resources but also actively engage in online learning experiences. The focus is on creating an intuitive and user-friendly interface to facilitate a seamless learning journey.
- **SUBJECT COVERAGE ACROSS ACADEMIC LEVELS:** The online learning platform aspires to be inclusive, offering a diverse array of subjects and courses tailored to cater to students across different academic levels. From Class 6 to class 10, the platform spans various academic disciplines, ensuring that learners have access to a broad spectrum of educational content relevant to their academic progression.
- **MOBILE ACCESSIBILITY:** Recognizing the ubiquity of mobile devices, the platform is designed for mobile accessibility. The mobile applications empower students to engage in learning activities at their convenience, breaking free from the constraints of a fixed physical location. This mobile-centric approach aligns with the modern lifestyle, allowing learning to seamlessly integrate into students' daily routines.
- **QUIZZES AND TESTS:** Assessing and reinforcing knowledge is integral to the learning process. The platform incorporates subject-based quizzes and tests that enable students to evaluate their understanding and proficiency. These interactive assessments not only serve as tools for gauging knowledge but also contribute to a more engaging and participatory learning experience.

- **SUBJECT SEGMENTATION:** To enhance navigability and organization, the platform employs subject segmentation. Content is categorized into different subjects, providing a structured and easily navigable framework for students. This segmentation streamlines the learning process, allowing students to locate and engage with specific subjects with ease.

In essence, the comprehensive online learning platform outlined here is not just a repository of educational materials but a dynamic and accessible educational ecosystem that caters to the diverse needs and preferences of students across academic levels.

Chapter 2: LITERATURE REVIEW

Numerous researches and studies have been conducted upon the effectiveness of various teaching-learning methods. Teaching-learning process has evolved drastically over the years. Modern teaching-learning process uses the strategy of Constructivism. Constructivism theorizes the fact that, learning is an individual activity. In the Constructivist model, teacher does not behave as a sage on stage, but acts as a facilitator of Learning. (Singh and Sangeeta, 2015).

In the constructive learning process students do not sit idly remain as passive gainers of knowledge but are active constructors of “knowledge through experience, observation, documentation, analysis and reflection,” Constructivist teachers encourage to constantly assess their understanding. Students in a constructivist classroom attain expertise in learning through the act of self-questioning and analyzing the strategies they undergo. (Bada and Steve, 2015).

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Dr Sunita Singh and Sangeeta Yaduvanshi (2015) in the paper, “Constructivism in science classroom: why and how,” propounded that “Constructivism is helpful in the learning of Science in true sense i.e., not only as a body of knowledge but also as a process for making sense of surroundings,” Students learning through constructive approach “did have some impact on their performance in mathematics in terms of their understanding and applicability and that they were able to 2013). Constructive techniques ensure that learn difficult subjects with effectiveness and ease. Constructivism enhances the students to be critical learners. Rote learning was a strategy of the past modern learners learns by understanding the concept.

Renu Yadav (2016) in "Role of Constructivism in Learning," opines that in the Constructive classroom, scaffolding helps the students to “develop the ability of reflection,” It enhances their capacity to develop” critical voice and shared vision, which helps to increase their academic

output and also helps them to develop leadership qualities and enhance “socio-emotional learning”.

In the modern education scenario, M-learning has a serious impact on the teaching-learning process. Adnan Majeed (2015) in the paper, "Survey Paper on Mobile Learning and Education," explicates that Mobile Learning transforms the entire educational scenario and it has lot of business and education applications. Students increasingly use smartphones, tablets, iPads, PDAs, online education apps to improve productivity in their studies.

M-learning can help both the students and teachers to solve the problems of “traditional learning system”, (Sarrab, Laila and Hamza, 2012). M- Learning access as a handy system that interact with the learner and teacher and facilitates teaching-learning process. Both teachers and students need a proper and handy system to inter- act with each other and facilitate the teaching-learning process. “M-learning systems are not to replace the traditional classrooms, but they can be used to complement the learning process in our schools and universities”, (Sarrab, Laila and Hamza,2012).

Rodovan, Vrana (2018) in” Acceptance of mobile technologies and M- learning in Higher education Learning: An Explorative Study at the Faculty of Humanities and Social Science at the University of Zagreb," says that mobile technology gets highly integrated into the personal and working environment of the students. He suggests that academic institutions should promote M-learning by incorporating it into their curriculum.

Ann Marie Casanova (2018) in her paper, "Case study - Cultivating a love of learning conceptual understanding that is Improving educational Outcomes in India (English)," elucidates that Biju’s app “wants to revolutionize the way millions of students think and learn” effectively. The app ensures that it cultivates a generation of learners who have a curiosity present in them to learn more, dream more, and become more.

2.1 EXISTING SYSTEMS

2.1.1 MeroSpark

MeroSpark is an educational website where students can get educational materials which can assist on their study. More than that, it is a community of learners, where knowledge is shared. Here you can ask the question, answer other's question and you can share your own articles. You can also buy second hands books on cheaper price or if you have any books which you are not using, you can sell it here such books. App is limited to interactive features and potential content gaps.

2.1.2 BYJU's

BYJU's – arguably the most popular online learning app in India. Although the app primarily focuses on school students, it also includes courses for competitive exams such as JEE/NEET, UPSC, and Bank PO. While the courses are paid – starting at Rs 3,500, there is a scholarship programmed for students in classes 4th-12th. BYJU's also include online classes for conceptual understanding, clearing doubts, personalized learning, and individual attention from mentors. Additionally, the company's recent investment in Whitehat Jr that teaches coding, which was recently included in India's new education system, to kids. But this app is limited to providing a separate or dedicated application for parent.

2.1.3 MeroSchool

Mero School is an online learning app that provides online video classes that tend to cover every base intended for school, high school and bachelor's level curriculum such as Engineering, Management, science. They also provide various language classes. App has limitation of buying course easily from the app.

2.2 LIMITATIONS OF PREVIOUS SYSTEMS

Although there are a couple of existing systems, some of the limitations include:

- Parents are not provided with real-time progress reports of their students' live performance.

- E-learning platforms like MeroSchool encounter challenges such as the absence of personalized learning features, making it difficult to cater to individual preferences.
- E-learning faces limitations with high latency.
- Best e-learning platforms come with a high price tag, making them financially challenging for a significant number of users.

2.3 SOLUTION PROPOSED BY OUR SYSTEM

To address above limitations, our system considers the following strategies:

- Our application also offers a parent version, providing a live progress report and showcasing the activities of the students.
- At the time of registration, students can choose their courses in advance, ensuring a personalized experience within the app.
- Our app minimizes latency through advanced technology and localized servers, optimizing the storage of data, including videos and notes.
- Our app efficiently handles computational tasks within the mobile application, ensuring secure storage of essential data on the device. This eliminates the reliance on external servers and their resources, contributing to cost reduction.

Chapter 3: METHODOLOGY

3.1 PROCESS MODEL

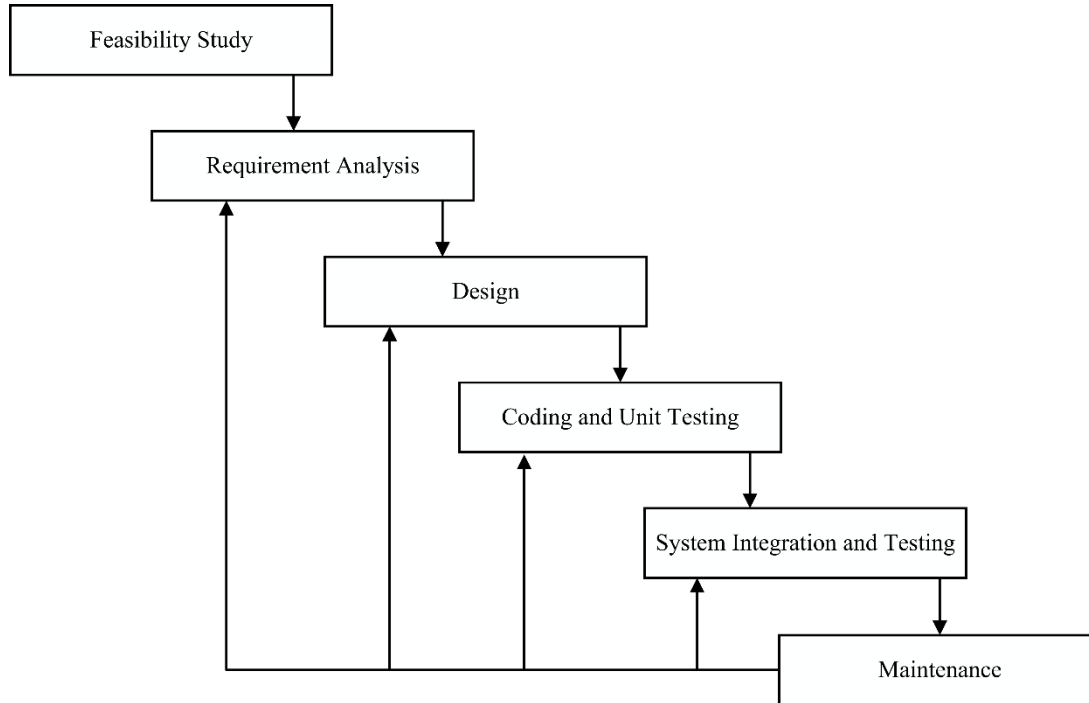


Figure 3.1: Iterative Waterfall Model

The iterative waterfall model is a software development approach that combines the sequential structure of the traditional waterfall model with the flexibility of iterative development. In this model, the development process is divided into several phases, including requirements gathering, design, development, testing, and deployment.

Unlike the traditional waterfall model, the iterative waterfall model allows for feedback and revisions. It follows a cyclical approach, where each iteration focuses on a specific set of requirements or features. The feedback paths allow the phase to be reworked in which errors are committed and these changes are reflected in the later phases.

For the development of our application, we have been utilizing the iterative waterfall model. This approach has been helping us to break down the project into manageable modules and progress through each phase iteratively.

In the feasibility analysis stage of our e-learning application project report, we investigated comparable projects. Based on recommendations from individuals at the Ministry of Education, Science and Technology, an app designed to facilitate a personalized and engaging learning experience could prove effective if executed properly.

During the planning phase, gathering and eliciting requirements emerge as crucial activities with a profound impact on the project. This involved comprehensive studies through literature review, interviews, and distribution of questionnaires among students from various universities in Nepal. Additionally, a review of applications with similar or desired features was conducted. The findings from these studies contributed to the finalization of a comprehensive list of features for implementation.

The project involved gathering requirements and scoping to analyze and design its workflow and architecture. Use case diagrams and detailed descriptions were created. Prototypes, including both high and low fidelity, were generated for stakeholder feedback. The low-fidelity prototype acted as a wireframe for user views and interfaces, analyzed for user experience.

In the integration and testing phase, the project components are combined and rigorously tested to ensure seamless functionality. This includes unit testing for individual modules, integration testing for component interactions, and system testing for overall compliance. Performance, security, user acceptance, regression, load, and stress testing are conducted to ensure a robust and reliable final product. Thorough documentation tracks issues, changes, and resolutions throughout this critical phase.

Applying the iterative waterfall model, our e-learning project advanced smoothly. The feasibility analysis, informed by comparable projects and insights from education authorities, established a strong foundation. Comprehensive requirement gathering in the planning phase outlined essential features. Design, with prototypes and stakeholder feedback, clarified the workflow. Rigorous integration testing ensured the final product's reliability. Emphasizing adaptability and continuous feedback, our approach positions the application to effectively deliver a personalized and engaging learning experience.

3.2 SYSTEM BLOCK DIAGRAM

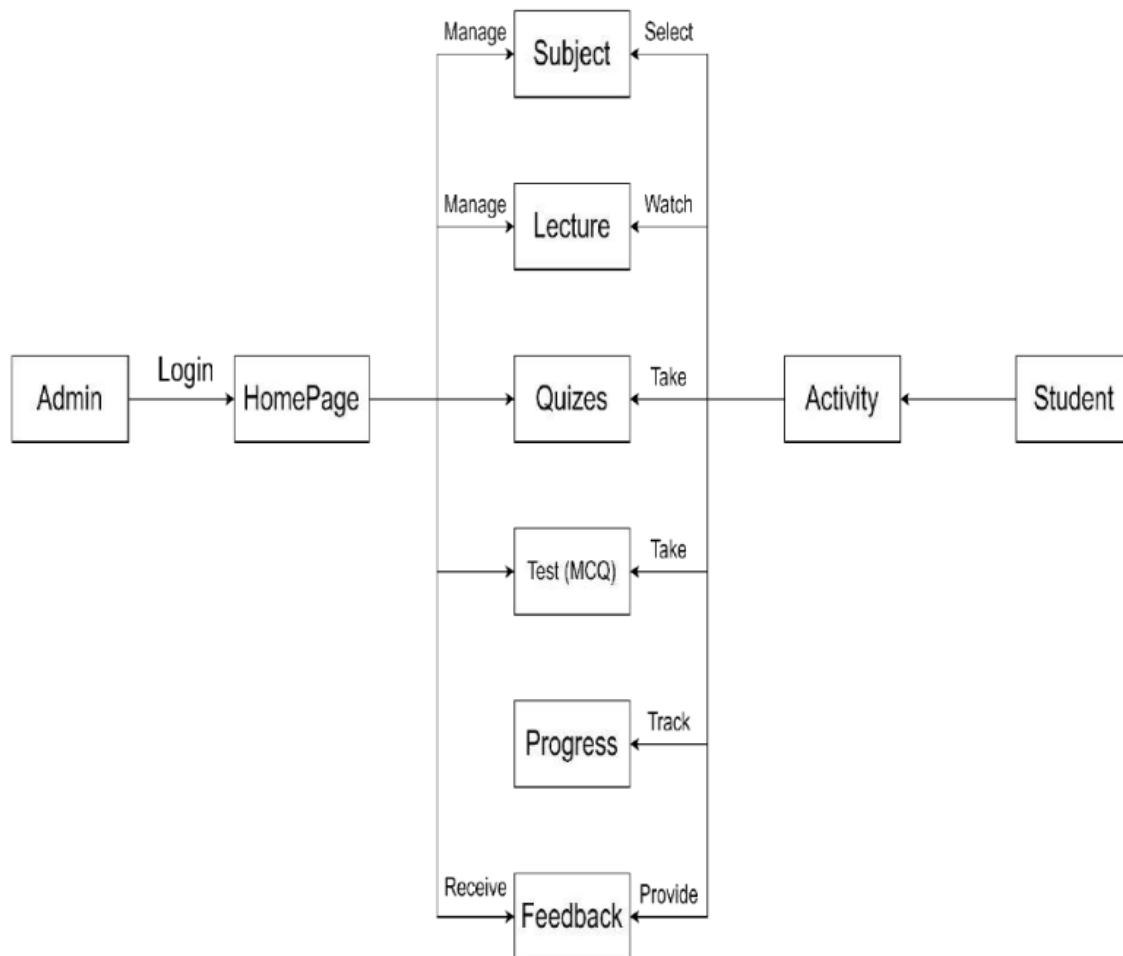


Figure 3.2: System Block Diagram

Students can easily sign up on the app using their phone number, email, and real name, making the registration process simple and accessible. Once registered, they have the flexibility to select and study the courses that align with their educational needs, and the app's homepage conveniently displays content related to their chosen courses. In the store section, students can explore and purchase packages, granting them access to premium, paid content tailored to their academic requirements. After acquiring a package, they can seamlessly transition to the study screen, where they can choose a specific subject and engage with lecture videos.

Within the study screen, students not only watch lecture videos but also have the option to enhance their understanding by taking notes directly within the app. These notes serve as valuable study aids, allowing students to revise and reinforce their knowledge at their own pace

also the app provides support for PDF notes, offering students a comprehensive tool to revise and reinforce their knowledge. For collaborative learning and addressing doubts, there's a dedicated discussion section where students can interact with each other. Additionally, if a student has a specific question for a teacher, they can utilize the "Ask a Doubt" option to receive personalized assistance.

To assess their comprehension, students can take chapter tests independently during their preferred study times. For a more competitive experience, the app offers periodic live tests, allowing students to gauge their knowledge on a global scale. Outstanding performance in these tests receives recognition on the Nepal leaderboard within the app. To cater to students with intermittent internet access, the app provides the option to download videos. These downloaded videos are securely stored using advanced technologies like DRM and AES encryption, mitigating the risk of piracy and ensuring a reliable offline learning experience.

3.3 FLOWCHART

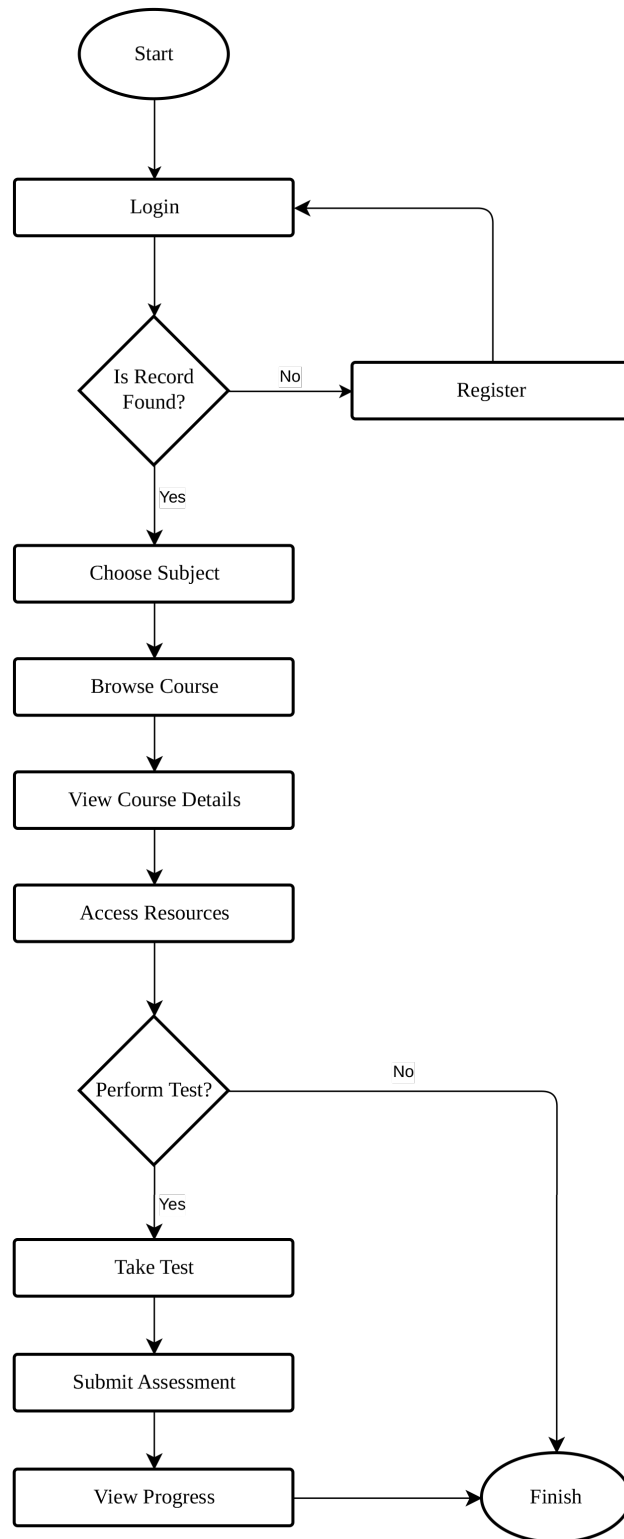


Figure 3.3: Flowchart of Application

3.4 USE CASE DIAGRAM

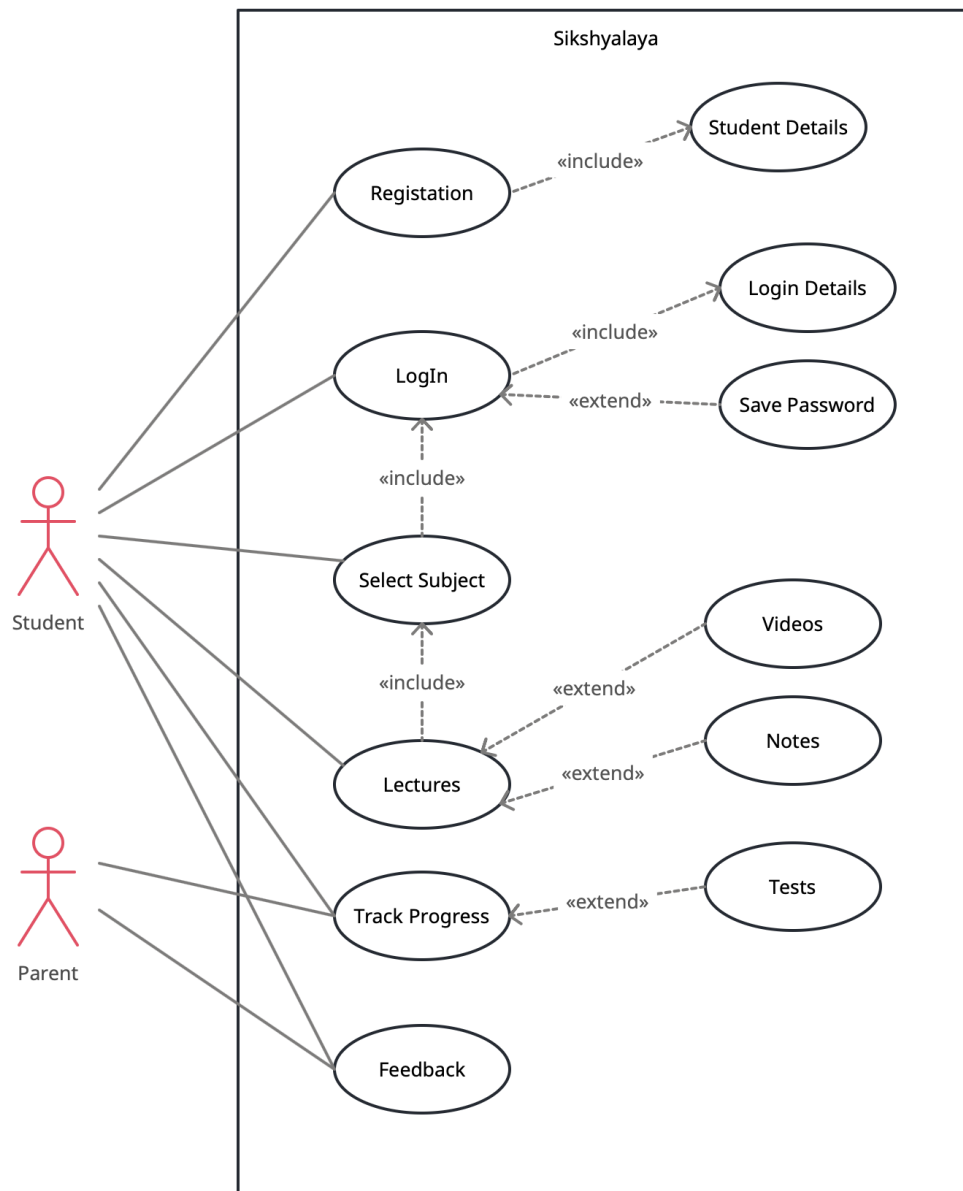


Figure 3.4: Use Case Diagram

3.5 OBJECT DIAGRAM

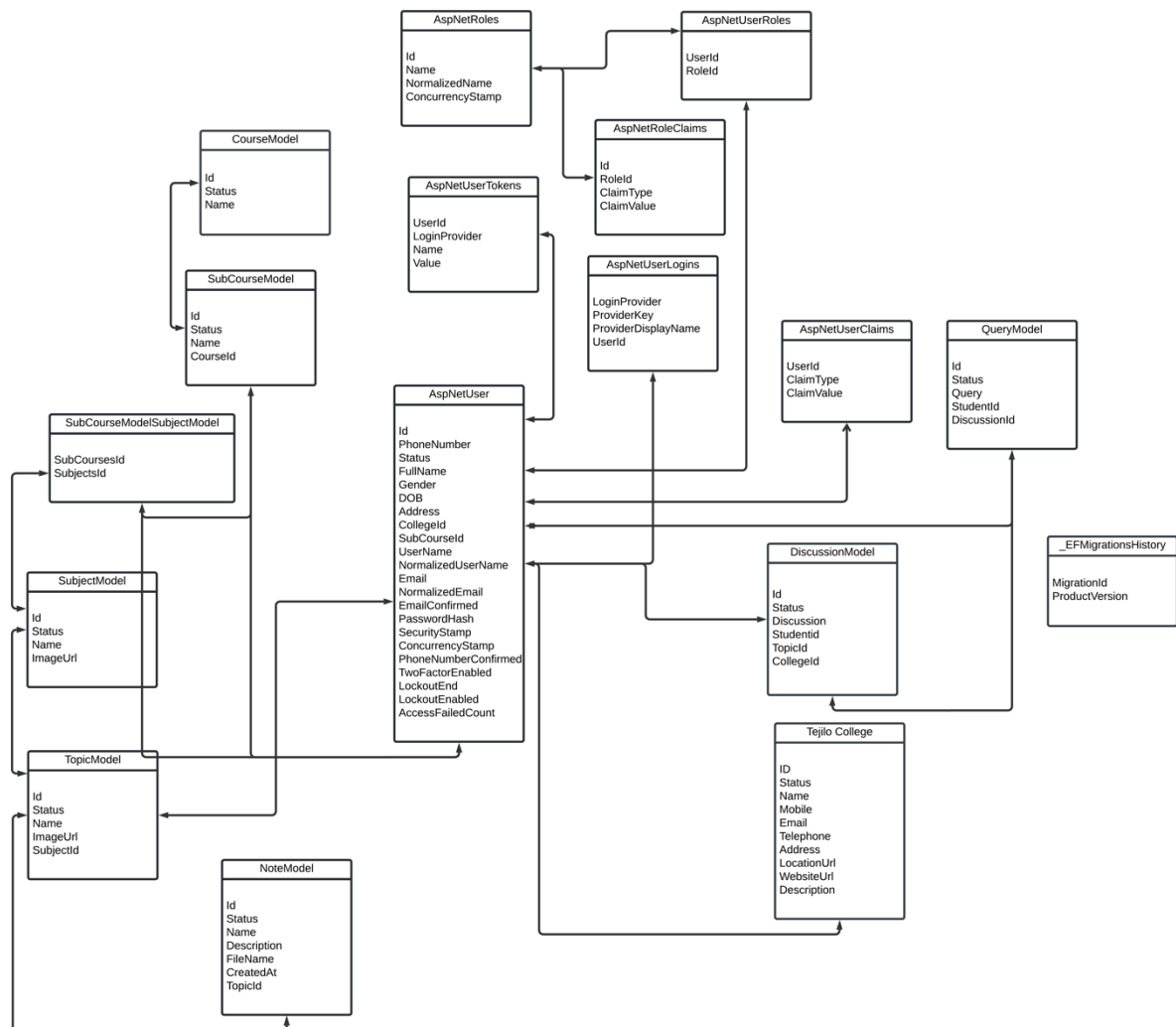
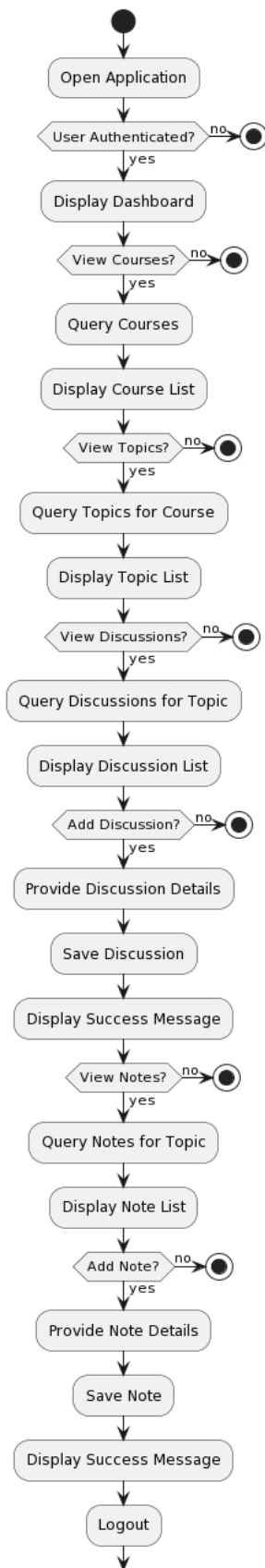


Figure 3.5: Object Diagram

3.6 ACTIVITY DIAGRAM



3.7 DETAILED SEQUENCE DIAGRAM

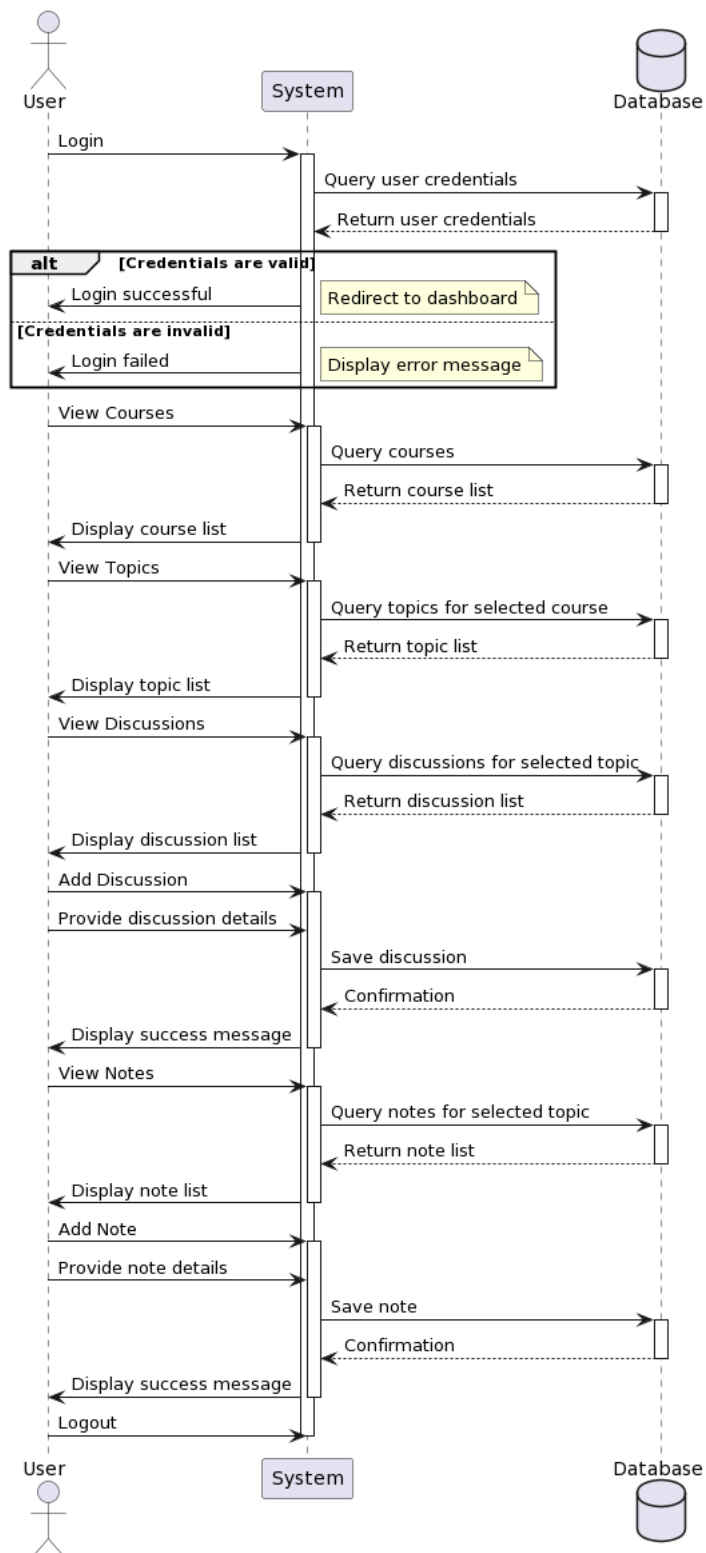


Figure 3.7: Detailed Sequence Diagram

3.8 DETAILED LOGIN SEQUENCE DIAGRAM

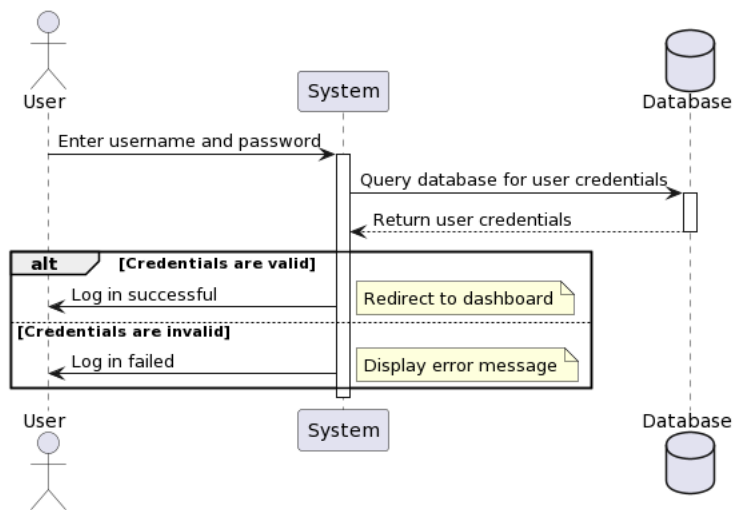


Figure 3.8: Detailed Login Sequence Diagram

3.9 DETAILED TOPIC MODEL SEQUENCE DIAGRAM

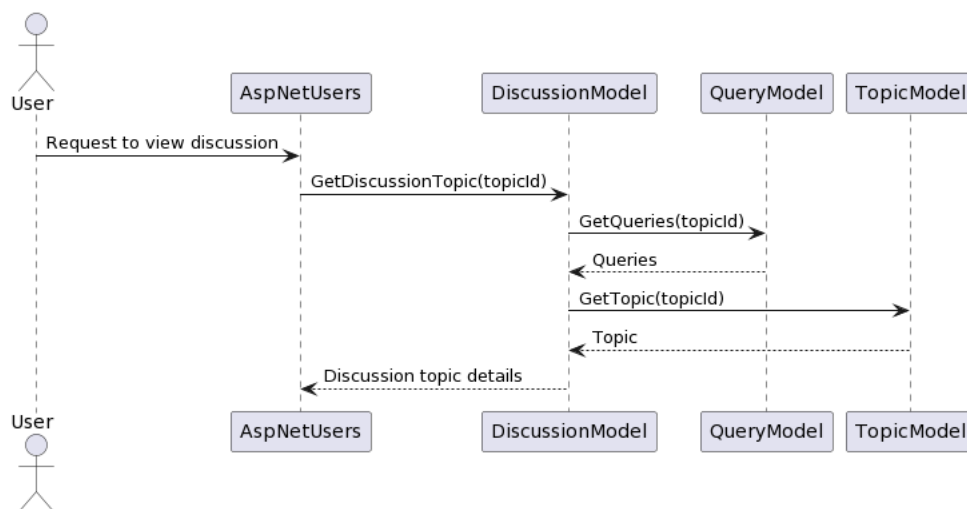


Figure 3.9: Detailed Topic Model Sequence Diagram

3.10 STATE DIAGRAM

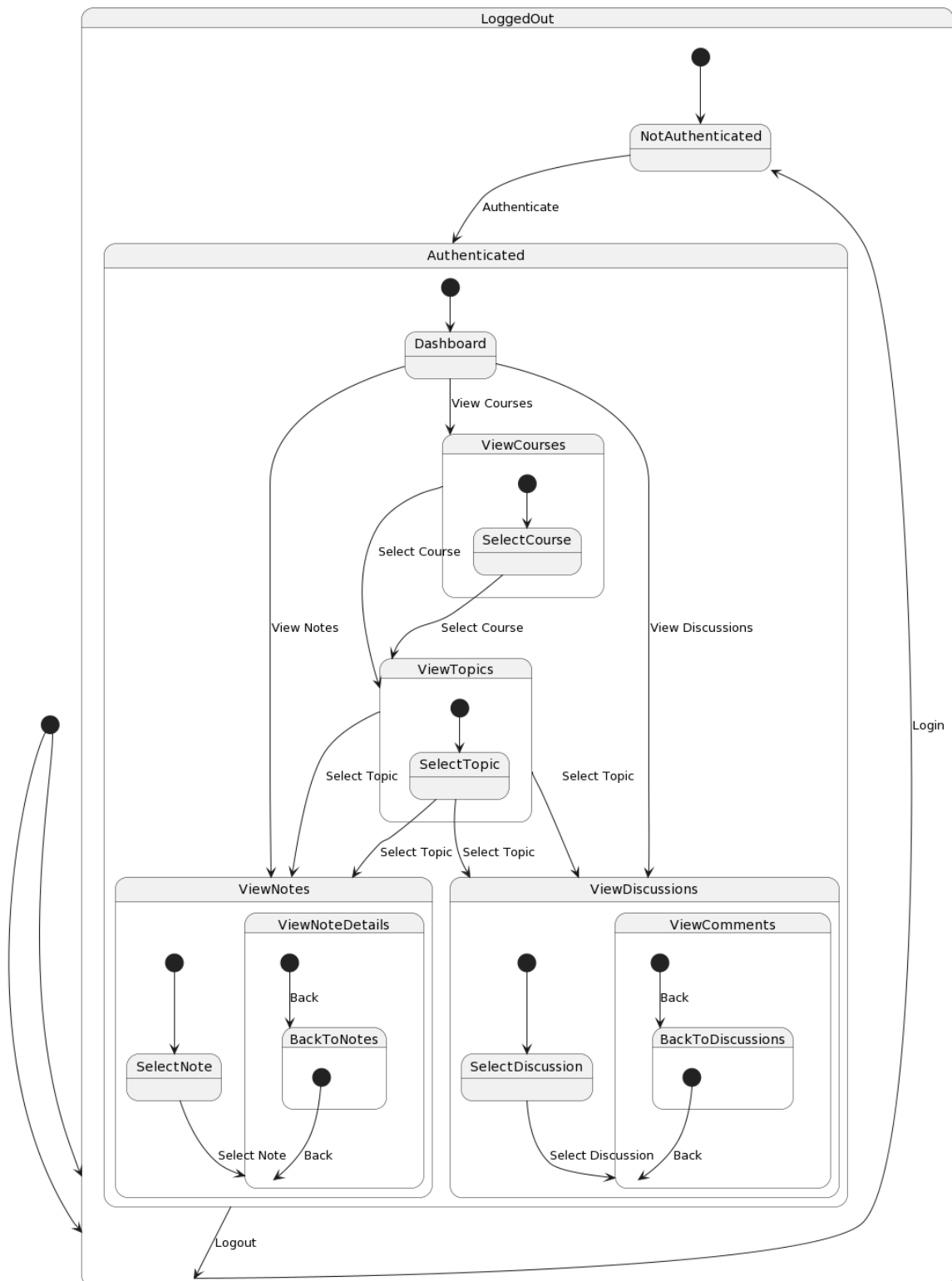


Figure 3.10: State Diagram

3.11 TOOLS USED

3.11.1 KOTLIN

Kotlin has rapidly gained popularity among developers due to its modern features and seamless integration with existing Java codebases. With its concise syntax and expressive language features, Kotlin significantly reduces the amount of boilerplate code required compared to Java, making codebases cleaner and more maintainable. One of its standout features is its comprehensive support for null safety, which helps prevent common runtime errors caused by null pointer exceptions. This feature alone has been a game-changer for many developers, allowing them to write safer code with fewer bugs.

Another key aspect of Kotlin is its support for asynchronous programming through coroutines. Coroutines enable developers to write asynchronous code in a sequential style, making asynchronous tasks easier to manage and understand compared to traditional callback-based approaches. This has greatly simplified the development of responsive and scalable applications, particularly in the context of Android development where asynchronous operations are common. By providing a more intuitive and efficient way to handle concurrency, Kotlin's coroutine support has been widely praised by developers.

Moreover, Kotlin's full interoperability with Java makes it an excellent choice for both new projects and existing codebases. Developers can seamlessly use Kotlin alongside their Java code, gradually migrating to Kotlin as needed without disrupting their existing workflows. This level of compatibility ensures a smooth transition for teams adopting Kotlin and allows them to leverage the language's modern features without sacrificing compatibility with existing libraries and frameworks. Overall, Kotlin's combination of modern language features, null safety, coroutine support, and interoperability with Java makes it a compelling choice for JVM and Android development, empowering developers to write safer, more concise, and more maintainable code.

3.11.2 ANDROID STUDIO

Android Studio is the official IDE for Android application development, based on Intel On top of the capabilities you expect from IntelliJ IDEA, Android Studio offers: Flexible Gradle-based build system, Code templates to help you build common app features, Rich layout editor with

support for drag and drop theme editing, 'Lint' tools to catch performance, usability, version compatibility.

3.11.3 ANGULAR

Angular is a powerful framework known for its ability to create interactive and dynamic web applications, making it an ideal choice for projects like Sikshalaya. By leveraging Angular for front-end development, Sikshalaya benefits from a responsive and intuitive user interface. Angular's robust features, such as two-way data binding, dependency injection, and component-based architecture, enable developers to build rich and engaging experiences for users. Whether it's handling complex data interactions or seamlessly updating the UI, Angular provides the tools necessary to craft modern web applications that meet the needs of today's users.

In the context of Sikshalaya, Angular's flexibility allows for the creation of a responsive design that adapts seamlessly across various devices and screen sizes. This ensures that users can access the platform from desktops, laptops, tablets, or smartphones without sacrificing usability or functionality. With Angular's support for reactive programming and its extensive library of pre-built components, developers can streamline development efforts and focus on delivering a polished user experience. Whether users are accessing Sikshalaya on-the-go or from the comfort of their homes, Angular empowers the platform to deliver consistent performance and usability across different environments.

Moreover, Angular's robust ecosystem, extensive documentation, and active community support further enhance the development process for Sikshalaya. Developers can leverage a wide range of third-party libraries, tools, and resources to expedite development and address specific project requirements. From managing state with NgRx to optimizing performance with Angular Universal, the ecosystem surrounding Angular offers solutions to common challenges faced during front-end development. This enables the Sikshalaya team to iterate quickly, incorporate user feedback, and continuously improve the platform to meet the evolving needs of its users.

3.11.4 DOT NET CORE WEBAPI

NET Core WebAPI is a powerful runtime environment that empowers developers to execute their C# or F# code on the server-side. It boasts an event-driven, non-blocking I/O model, making it adept at managing concurrent operations and handling network requests efficiently. In the Sikshalaya project, .NET Core WebAPI serves as the backbone of the backend infrastructure, offering scalability and robustness for implementing server-side logic and APIs. With .NET Core WebAPI, developers can leverage the extensive capabilities of the .NET ecosystem, including libraries and frameworks, to build resilient and high-performance web applications. Its asynchronous programming model allows for seamless handling of multiple requests simultaneously, ensuring optimal resource utilization and responsiveness. In Sikshalaya, .NET Core WebAPI plays a pivotal role in delivering a reliable and efficient platform for serving dynamic content and interacting with clients.

By utilizing .NET Core WebAPI in the Sikshalaya project, developers can harness the versatility and reliability of the .NET runtime for building scalable and performant backend services. Whether managing user authentication, processing data, or handling HTTP requests, .NET Core WebAPI provides the necessary tools and infrastructure to streamline development and deployment workflows. With its emphasis on performance and scalability, .NET Core WebAPI stands as a robust choice for powering the server-side components of Sikshalaya and ensuring a seamless user experience.

3.11.5 SWIFT

Swift, Apple's programming language designed for iOS, macOS, watchOS, and tvOS app development, stands out for its native integration and optimized performance within the Apple ecosystem. Created with a focus on modern syntax and safety, Swift offers developers a robust toolkit for crafting intuitive and efficient applications for various Apple devices. Its seamless interoperability with Objective-C allows developers to leverage existing codebases while embracing the advancements and improvements brought by Swift, ensuring a smooth transition and enhanced development experience.

One of Swift's key strengths lies in its performance capabilities, enabling developers to build high-performance applications that deliver a fluid user experience across Apple's range of devices. With features like automatic memory management and advanced compiler

optimizations, Swift optimizes code execution, resulting in faster app responsiveness and reduced resource consumption. This emphasis on performance not only enhances the user experience but also empowers developers to create feature-rich applications without compromising on speed or efficiency.

Moreover, Swift's ongoing evolution through regular updates and enhancements underscores Apple's commitment to providing developers with cutting-edge tools and technologies. By prioritizing simplicity, safety, and performance, Swift continues to empower developers to innovate and push the boundaries of app development within the Apple ecosystem. With its growing popularity and robust community support, Swift remains at the forefront of modern programming languages, driving the creation of immersive and impactful experiences for millions of users worldwide.

Chapter 4: EPILLOGUE

4.1 TASK COMPLETED

- Node.js has been utilized to complete the backend development.
- The process for purchasing the package has been successfully implemented.
- The functionality enabling PDF reading in the mobile app has been accomplished.
- Functionality to take MCQ tests within the mobile app.

4.2 TASK REMAINING

- To implement the functionality for uploading and streaming lectures.
- Development of an administrative panel website to manage the app's backend and overall functionality.

4.3 GANTT CHART

S. N.	TASK	2023						2024		
		JUL	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR
1	Documentation									
2	Feasibility Study									
3	Requirement Analysis									
4	Design									
5	Coding and Unit Testing									
6	System Integration and Testing									
7	Maintenance									

Figure 4.3: Gantt Chart

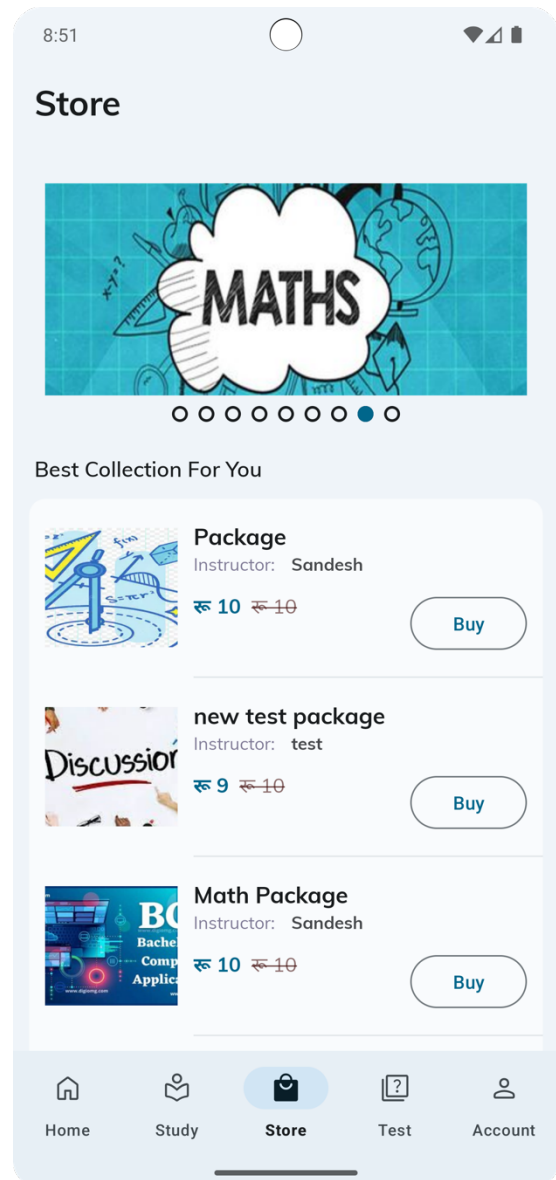
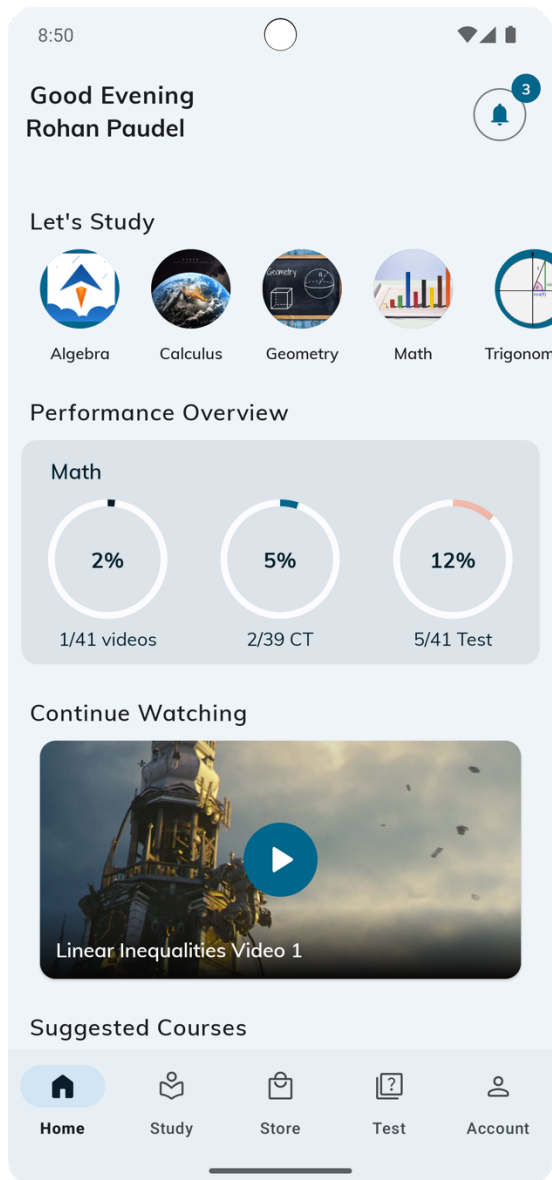
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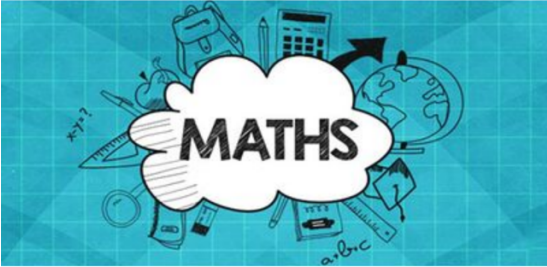
- [1] <https://unacademy.com>
- [2] <https://byjus.com>
- [3] https://www.researchgate.net/publication/342901964_Byju%27s_The_Learning_App_An_Investigative_Study_On_The_Transformation_From_Traditional_Learning_To_Technology_Based_Personalized_Learning
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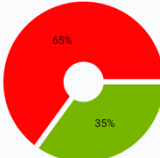
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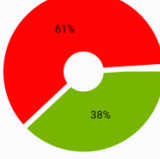
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